

Health and Wellbeing Trends in Global Cities: A Data-Driven Analysis of Paris's Performance

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Definition of the Business Problem

Paris is one of the countries that can be called the world's capital in terms of culture, art, and modernity. However, research on the healthy living index has revealed that people living in Paris have a rather low health score.

The aim of this project is to analyze the data presented to us and develop solutions to make Paris a more livable city.



Data and Analysis Methodology

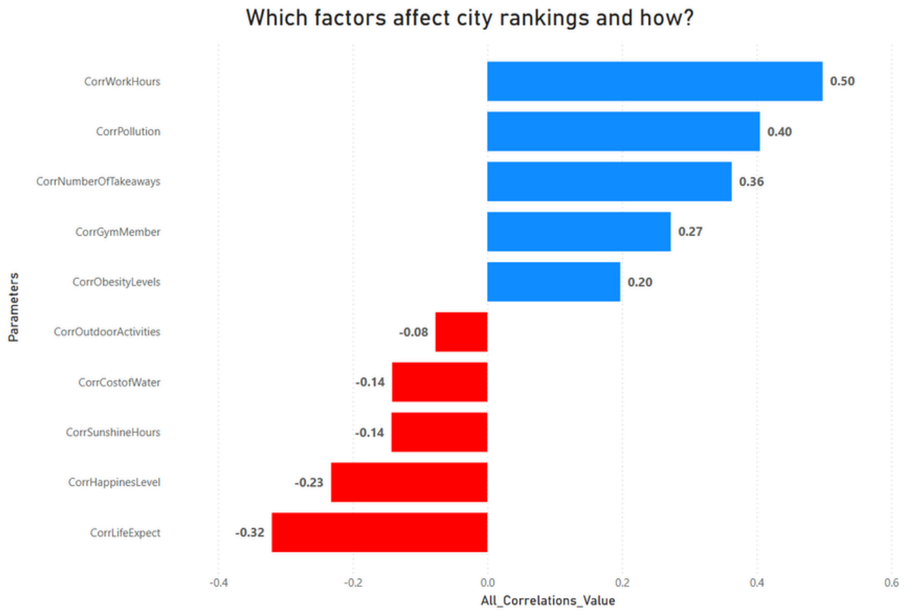
In this project, 2 types of analyzing methodology has been applied

1- Descriptive Statistics

Descriptive statistics were used to understand how well or poorly the data for the city of Paris performed compared to the average values of the relevant parameters.

Summart Statistics and Comparison

Parameters	Mean	Median	Min	Max	Paris	Comparison With Average
Sunshine hours	2245.9	2066	1405	3542	1662	⇩
Cost of a bottle of water	1.17	1.195	0.15	3.2	1.95	⇧
Obesity levels	22%	22%	4%	36%	21.60%	⇩
Life expectancy (years)	78.18	80.4	56.3	83.2	81.8	⇧
Pollution(Index score)	51.12	52.64	13.08	91.74	65.1	⇧
Annual avg. hours worked	1673	1686	1380	2137	1505	⇩
Happiness levels	6.44	6.9	3.57	7.8	6.66	⇧
Outdoor activities	214.0	189.5	23	585	331	⇧
Number of take-aways	1443	998	250	6417	4363	⇧
Cost of monthly gym membership	40.42	37.33	16.07	73.11	35.93	⇩

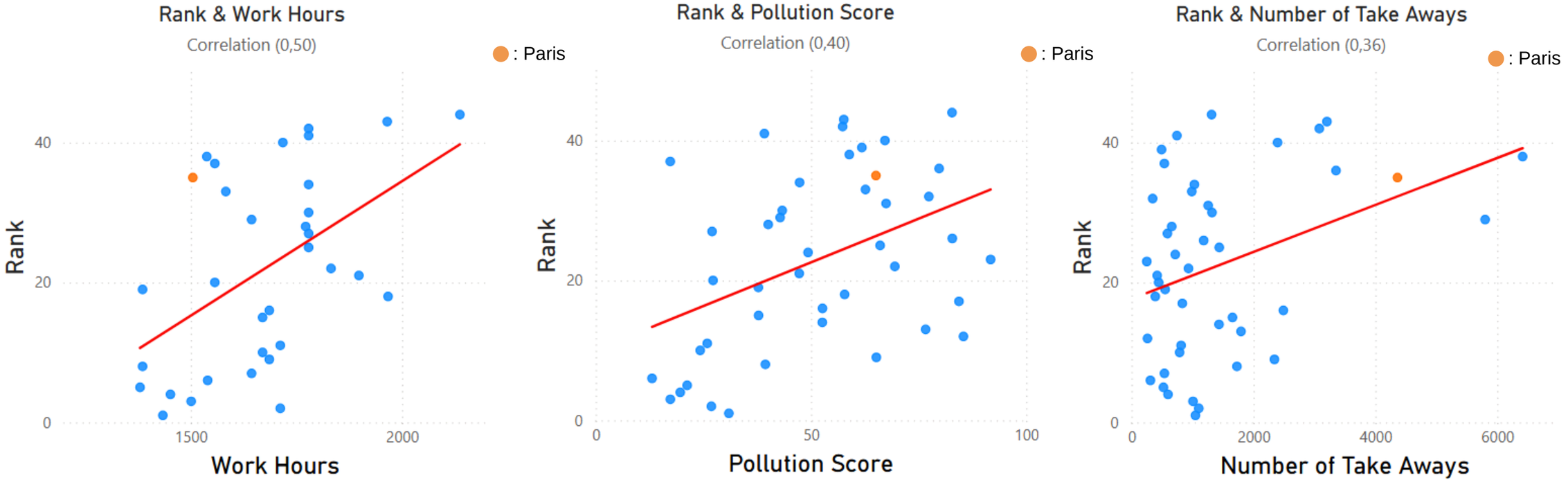


1- Correlation Analysis

Correlation analysis helps determine the extent to which parameters influence the well-being ranking, and whether this influence is positive or negative.

Parameters with a positive correlation are directly proportional to the ranking, while parameters with a negative correlation are inversely proportional.

Factors influencing the Well-Being Ranking - Scatter Analysis

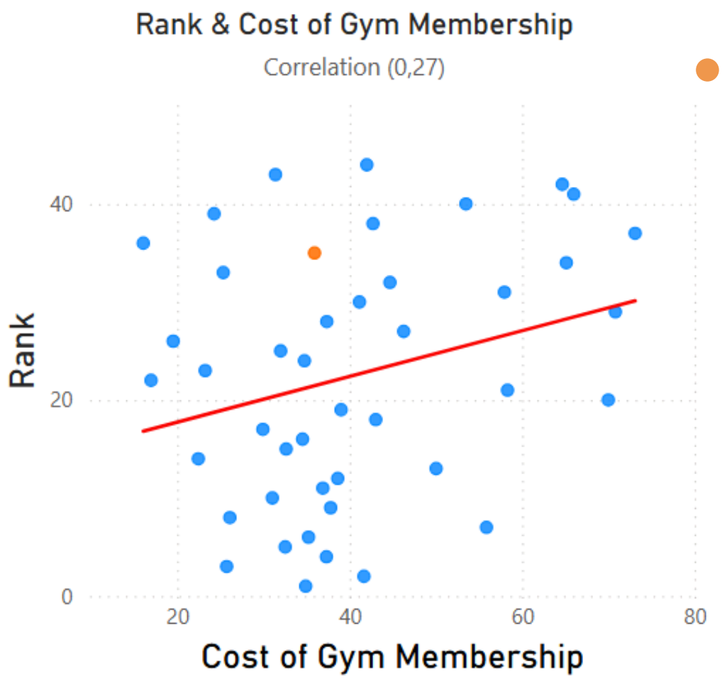


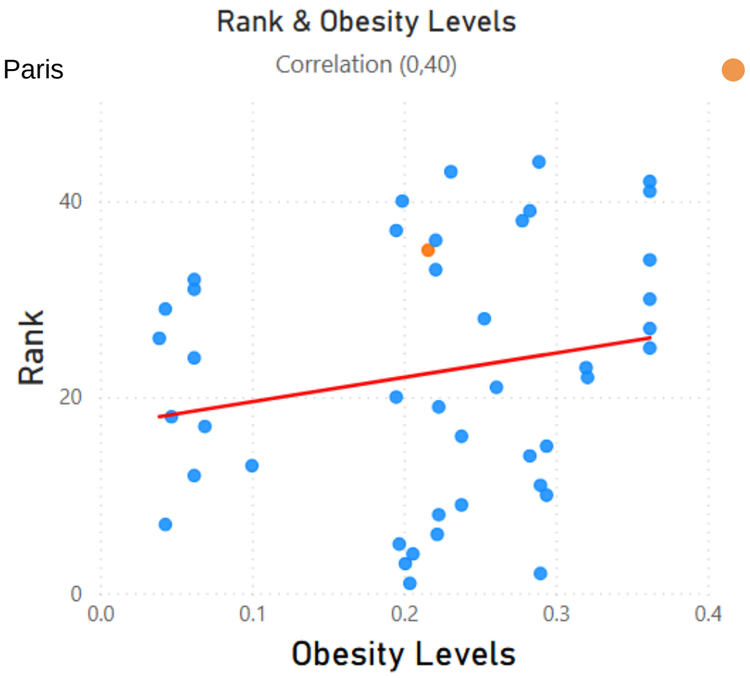
- **Work Hours & Rank:** The Work Hours parameter has the highest positive correlation with ranking. This means that longer working hours in a city can lead to a higher ranking, causing that city to fall lower in the well-being ranking.

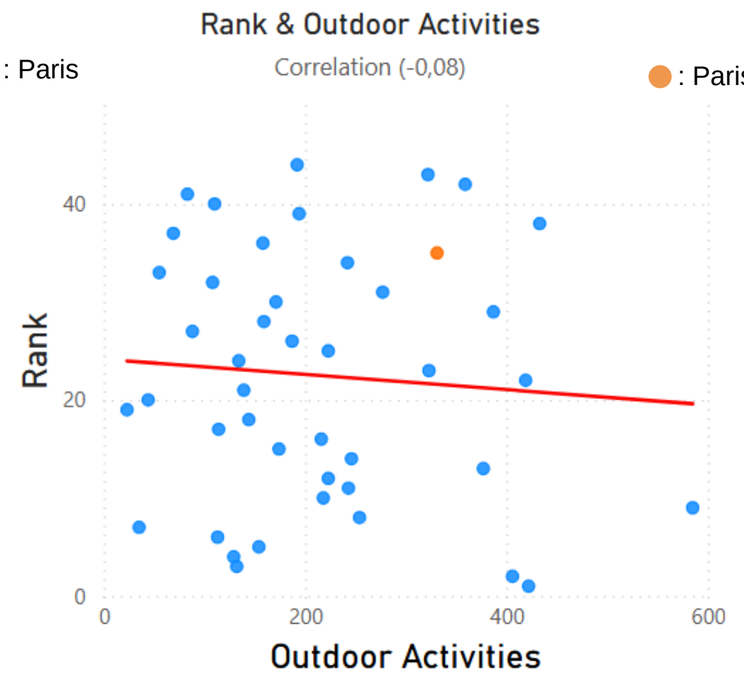
- **Pollution Score & Rank:** According to the correlation between Pollution Score and Rank, we observe that as the pollution level in a city increases, that city falls into worse rankings.

- **Number of Take Aways & Rank:** When we examine the correlation analysis between Number of Takeaways and Rank, we observe a positive correlation between these two parameters. This means that as the number of takeaway points increases in a city, it leads to a worse ranking due to fast-paced consumption and unhealthy lifestyle habits.

Factors influencing the Well-Being Ranking - Scatter Analysis





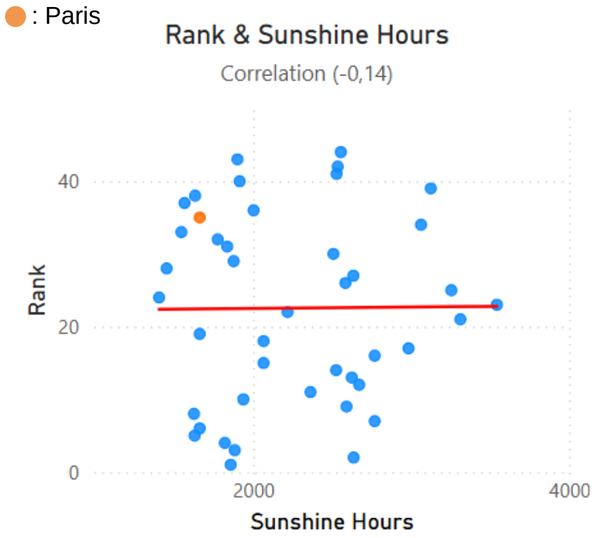


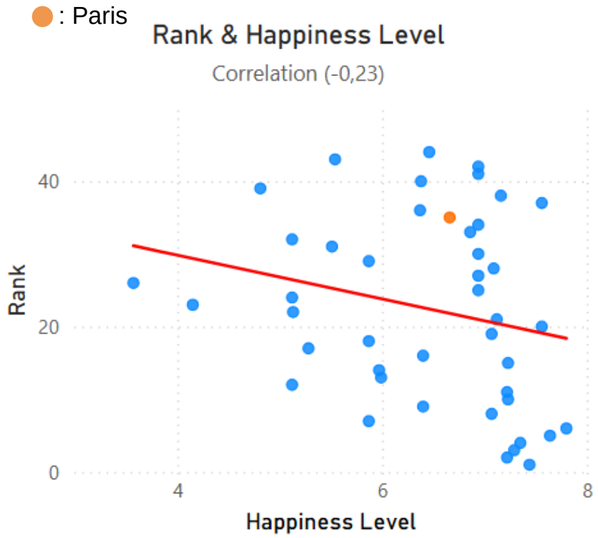
- **Cost of Gym Membership & Rank:** According to this correlation analysis, as the economic cost of exercising increases, the well-being ranking also increases. Consequently, this high cost negatively affects the ranking.

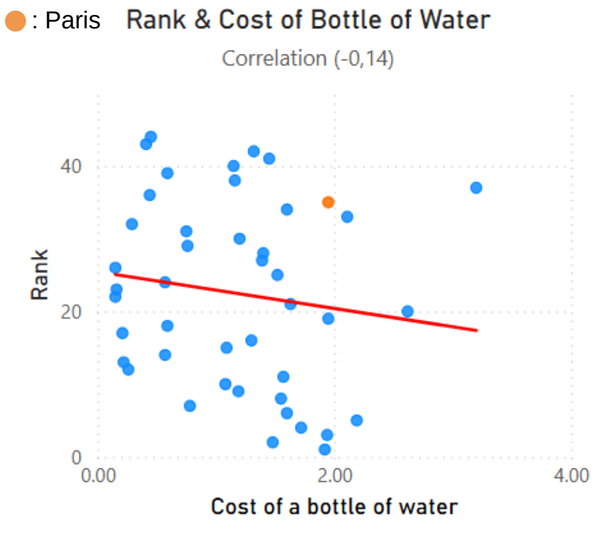
- **Obesity Levels & Rank:** According to this correlation, it has been observed that as the obesity rate in a city increases, its well-being ranking also increases proportionally, causing it to fall to lower ranks..

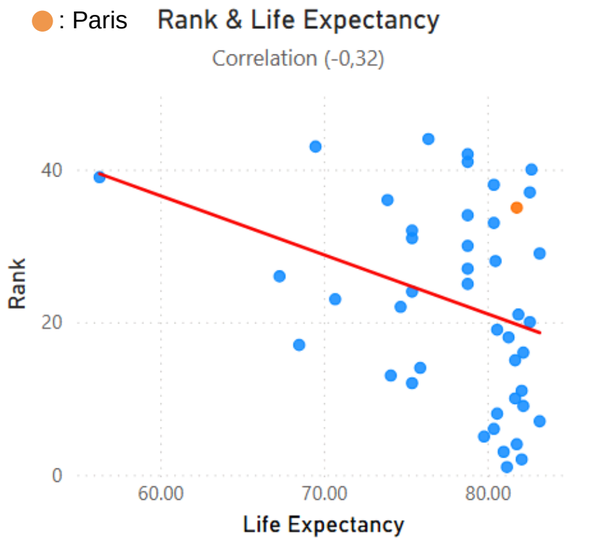
- **Outdoor Activities & Rank:** There is a negative correlation between the number of outdoor activities and ranking. This means that as the number of outdoor activities increases in a city, its well-being ranking decreases, leading to a better overall ranking for that city.

Factors influencing the Well-Being Ranking - Scatter Analysis



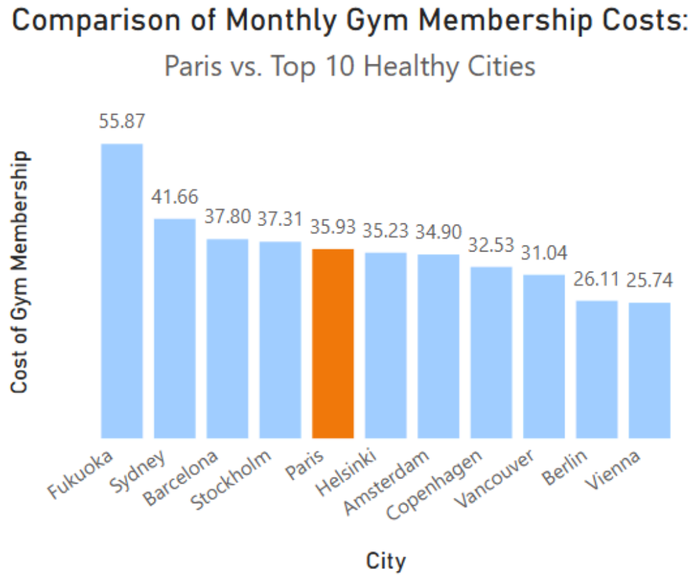


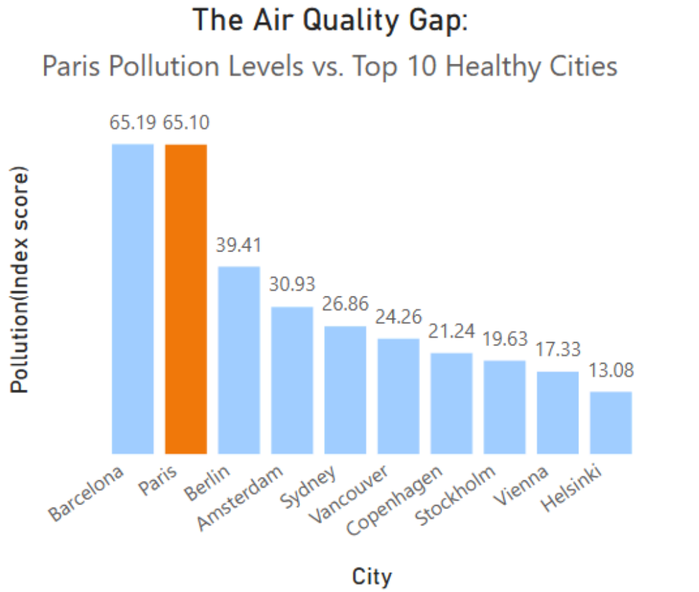


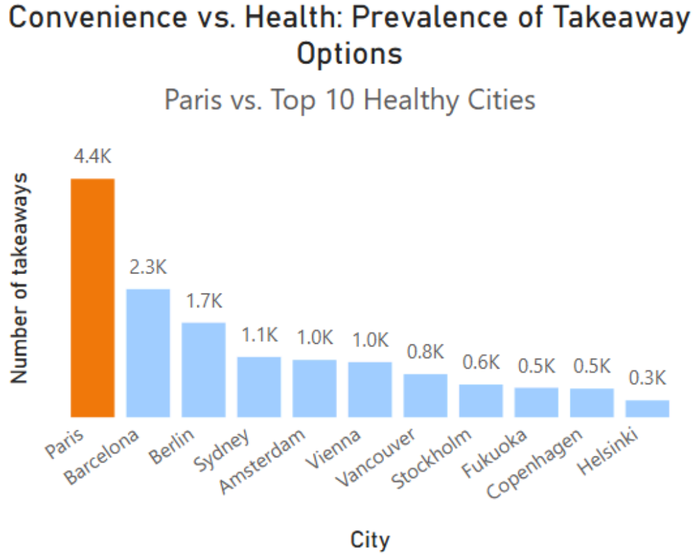


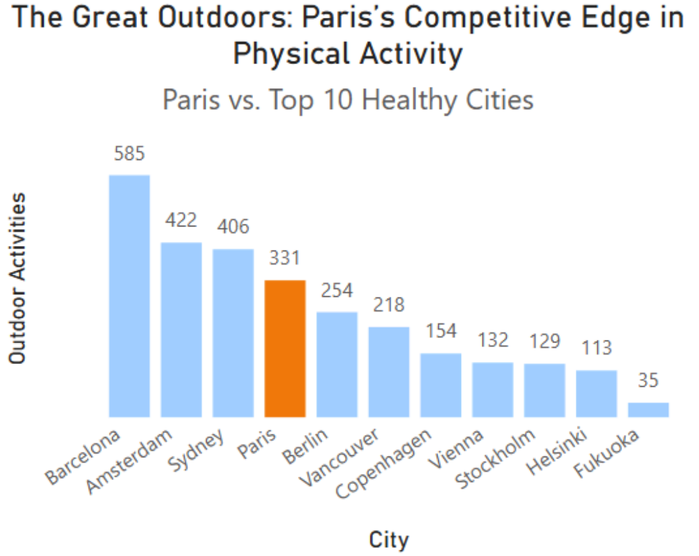
- **Cost of Bottle of Water & Rank:** There is a negative correlation between the cost of a bottle of water and ranking. This means that if the price of water is high in a city, that city will have a lower ranking, meaning a better ranking overall.
- **Sunshine Hours & Rank:** As the number of sunny days increases, the ranking decreases. This indicates that cities with a high number of sunny days have a better ranking.
- **Happiness Level & Rank:** There is a strong negative correlation between happiness levels and ranking. As the happiness rate of citizens increases, the city tends to have a better ranking.
- **Life Expectacy & Rank :** The correlation analysis revealed that as life expectancy increased, the ranking decreased, meaning it improved.

Factors influencing the Well-Being Ranking - Descriptive Analysis









- **Cost of Gym Membership:** When we compare gym costs in Paris with the top 10 cities, we observe that prices are above the average of these top 10 cities. The fact that gym costs are much lower in Vienna and Berlin is one of the main reasons why these two cities are in a better position than Paris.

- **Pollution Index Score:** The city of Paris, with a pollution score of 65.10, is almost 4 or 5 times higher than the healthiest cities, especially Helsinki and Vienna. This high pollution level is a significant factor that pushes Paris down in the rankings.

- **Number of Takeaways:** Paris has over 4400 takeaway locations. This number is at least 3 to 4 times higher than the average of the top 10 cities. This means that the incentive for unhealthy eating is quite high in the city.

- **Outdoor Activities :** When we examine outdoor activities, it ranks quite well among the top 10 cities. However, outdoor activities could be more.

Conclusion and Recommendation



We can summarize the analyses and recommendations we conducted in this section in three points:

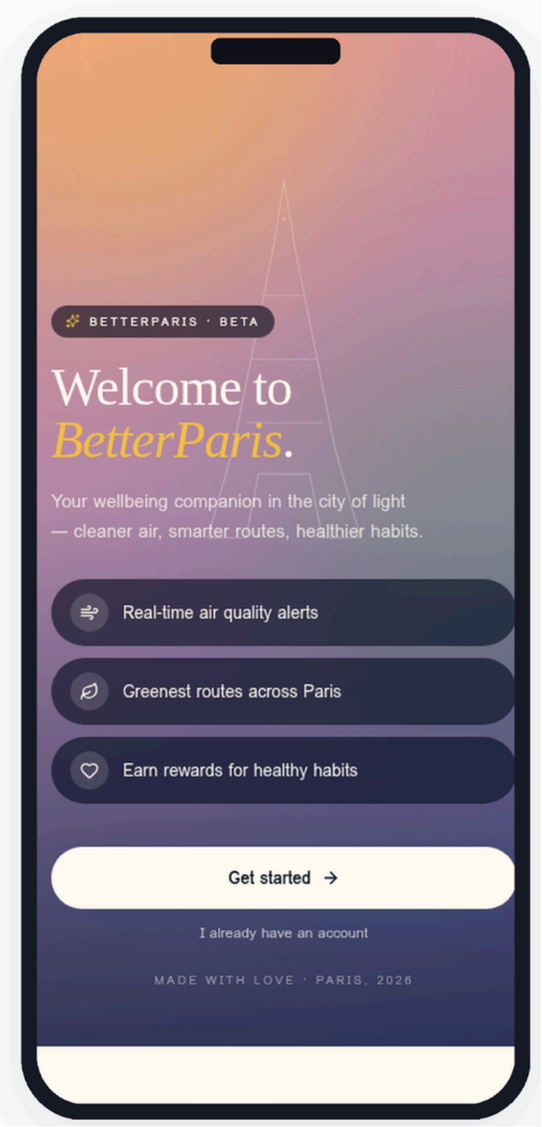
- Although Paris ranks quite well among the top 10 cities for outdoor activities, its high pollution score acts as a limiting factor.The pollution index is not a parameter we can control directly, but indirectly reducing it would improve the ranking of Paris.

- Paris's cost of gym membership is in the middle of the top 10 healthiest cities data. Since the cost of gym membership has a high correlation with the ranking and is a manageable parameter, reducing costs in this area will positively impact the ranking.

- The number of takeaways parameter provides us with very important information about the lifestyles of people living in Paris. The presence of 4400 takeaway points in the city, as a result of metropolitan life, leads people to consume fast and processed foods due to fatigue. This also negatively affects the city of Paris, depending on the correlation between the number of takeaway points and rank.

To improve Paris's ranking in the well-being index and enhance the quality of life for its residents, an interactive application called **BetterParis App** could be developed.

Implementation of Recommendations - BetterParis App



To improve Paris's ranking in the well-being index, we will dynamically collect and track the following 6 data sets:

Environmental Factor Data: By accessing BetterParis app users' location information, we will analyze their locations during peak pollution hours.

Nutritional Metrics: We will recommend healthy eating spots to users and compare this to the time they spend at takeaway points to help them improve their eating habits.

Gym Incentive Metrics: To overcome the cost barrier at gyms, we will dynamically track visits to the reward-based "pay-to-use" system integrated with the municipality.

Socialization Index: To improve the happiness score, we will track users' participation in community events and their interaction rates.

Work Hour Productivity: To increase the productivity of users' time spent in the office, we will track the ratio of time spent in the office to "Digital Detox" breaks. This will allow for planning to reduce weekly working hours in the long term.

Life Expectation Score: We will track users' cumulative life expectancy scores.

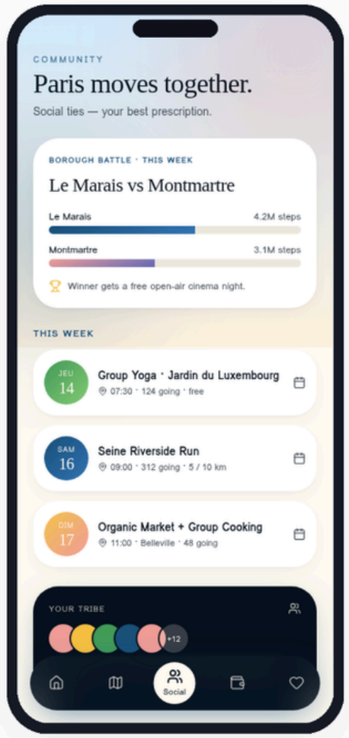
Flow Prototype - BetterParis App



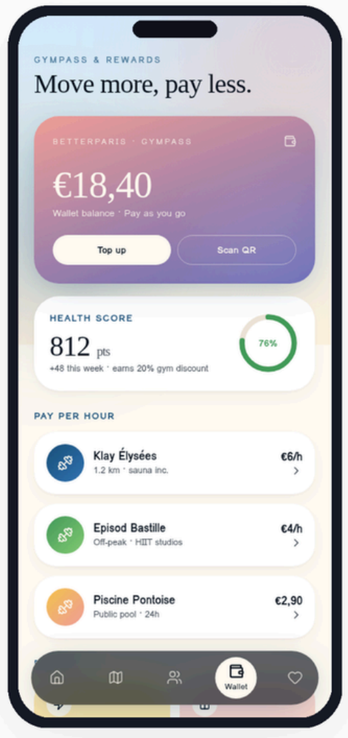
02 Smart Dashboard
Air quality + sun tracking



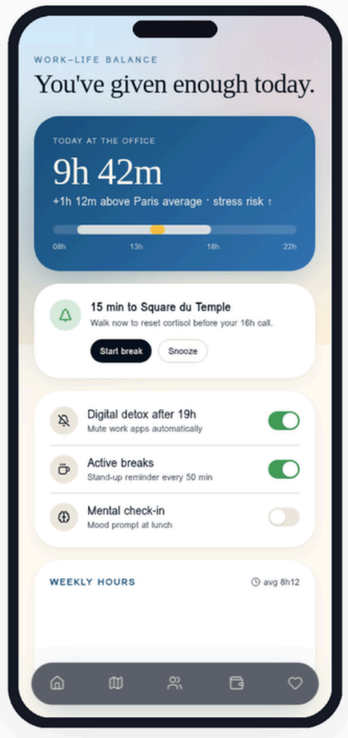
03 Healthy Map
Green routes + water + eats



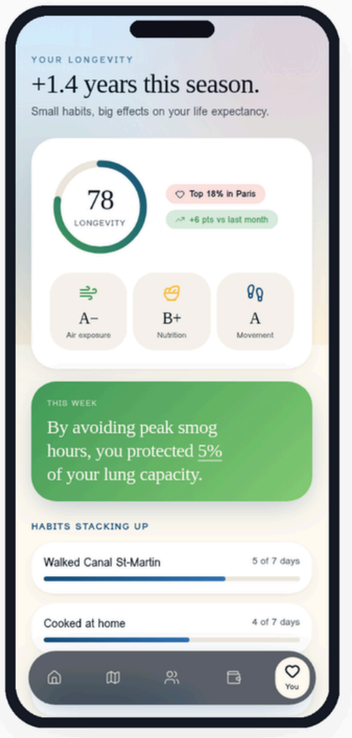
04 Community
Borough battles + events



05 Gym & Rewards
Pay-as-you-go + health score



06 Work–Life Balance
Office hours + active breaks



07 Longevity Score
Personalised wellbeing index

THANK YOU FOR BEING A PART OF BETTER PARIS

